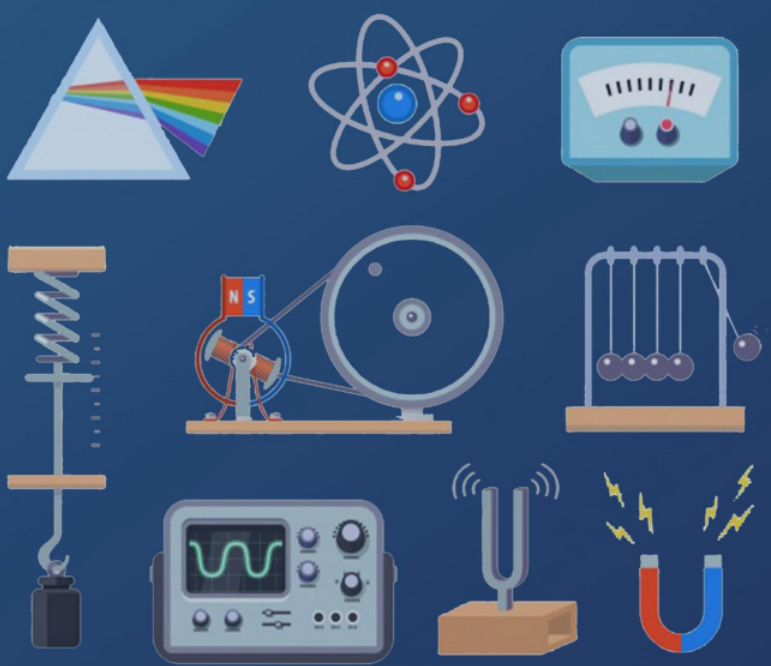
PHYSICS

Sound Wave



SHEET

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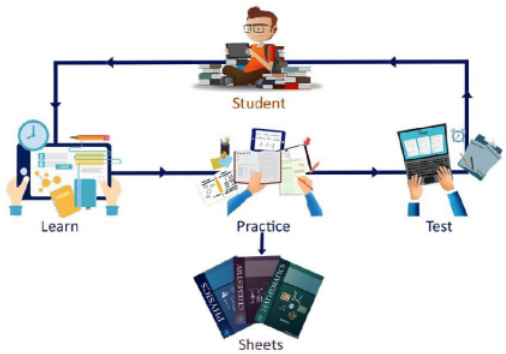
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to sound wave				
2	Equation of sound wave				1-3
3	Speed of sound wave	1-3	1,7,10	1-5	
4	Intensity of sound				4-6
5	Standing wave part 1				
6	Standing wave part 2	4-6	3	6-9	10-12,15
7	Resonance tube experiment	7-9	11	10-12	7
8	Doppler's effect	10-17			
9	Doppler's effect formula		4,12	13,14	13,14
10	Applications of Doppler's Effect	18	13	15-21	16,17
11	Interference	19-21	5,6,14	22	8
12	Beats	22,23	2,15	23-28	9

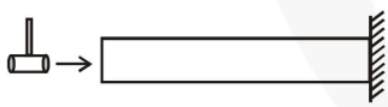
Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Speed of sound wave

Q1. A 1000 m long rod of density 10.0×10^4 kg/m^3 and having young's modulus Y = 10^{11} Pa, is clamped at one end. It is hammered at the other free end as shown in the figure. The longitudinal pulse goes to right end, gets reflected and again returns to the left end. How much time (in sec) the pulse take to go back to initial point?



(A) 4 s (B) 6 s
(C) 2 s (D) 10 s

Q2. At the room temperature the velocity of sound in O_2 gas is V. Then in mixture of H_2 and O_2 gas the speed of sound at same temprature:

(A) will be less than V
(B) will be more than V
(C) will be equal to V
(D) nothing can be said

Q3. Two monatomic ideal gases 1 and 2 of molecular masses m_1 and m_2 respectively are enclosed in separate container kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is given by :

(A) $\sqrt{\frac{m_1}{m_2}}$ (B) $\sqrt{\frac{m_2}{m_1}}$

(C) $\frac{m_1}{m_2}$ (D) $\frac{m_2}{m_1}$

Standing wave part-2

Q4. If the fundamental frequency for a Closed Organ Pipe is n, then the next three overtones will have ratio :-

(A) 2 : 3 : 4
(B) 3 : 4 : 5
(C) 3 : 7 : 11
(D) 3 : 5 : 7

Q5. A tube, closed at one end and containing air, produces, when excited, the fundamental note of frequency 512 Hz. If the tube is open at both ends,the fundamental frequency that can be excited is (in Hz).

(A) 1024 (B) 512
(C) 256 (D) 128

Q6. The velocity of sound in air is 330 m/s the fundamental frequency of an organ pipe open at both ends and length 0.3 metre will be :

(A) 200 Hz (B) 550 Hz
(C) 300 Hz (D) 275 Hz

Resonance tube experiment

Q7. The end correction of resonance tube is 1 cm. If lowest resonant length is 15 cm then next resonant lenght will be :-

(A) 36 cm (B) 45 cm
(C) 46 cm (D) 47 cm

Q8. An air column in pipe,which is closed at one end will be in resonance with a vibrating tuning fork of frequency 264 Hz if the length of the column in cm is : [v = 330 m/s]

(A) 31.25 (B) 62.50
(C) 110 (D) 125

Q9. If l_1 and l_2 are the lengths of air column for the first and second resonance when a tuning fork of frequency n is sounded on a resonance tube, then the distance of the displacement antinode from the top end of the resonance tube is :

(A) $2(l_2 - l_1)$ (B) $\frac{l}{2}(2l_1 - l_2)$

(C) $\frac{l_2 - 3l_1}{2}$ (D) $\frac{l_2 - l_1}{2}$

Doppler's effect formula

Q10. A siren blown in workshop emits waves of frequency 1000 Hz. A car driver approaches the workshop with velocity 90 km/hour then frequency of sound heard by driver will be in Hz. ($v_{sound} = 330\,m/s$)

(A) 926 (B) 1076

(C) 1176 (D) 1000

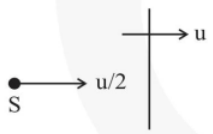
Q11. A wall is moving with velocity u and a source of sound moves with velocity u/2 in the same direction as shown in the figure. Assuming that the sound travels with velocity 10u, the ratio of incident sound wavelength on the wall to the reflected sound wavelength by the wall is equal to

(A) 9 : 11

(B) 11 : 9

(C) 4 : 5

(D) 5 : 4



Q12. A source of sound of frequency n and a listener approach each other with a velocity equal to $\frac{1}{20}$ of velocity of sound. The apparent frequency heard by the listener is:

(A) $\left(\frac{21}{19}\right)n$ (B) $\left(\frac{20}{21}\right)n$

(C) $\left(\frac{21}{20}\right)n$ (D) $\left(\frac{19}{20}\right)n$

Q13. A person observes a change of 2.5% in frequency of sound of horn of a car. If the car is approaching toward the person & sound velocity is 320 m/sec., then velocity of car in m/sec will be approximately.

(A) 8 (B) 800

(C)7 (D) 6

Q14. Two sound sources (of same frequency) are placed at distance of 100 meter. An observer when moving between both sources hears 4 beats per second. The distance between sound source is now changed to 400 meter then the beats/second heard by observer will be :

(A) 2 (B) 4

(C) 8 (D) 16

Q15. A railway engine moving with a speed of 60 m/sec passes in front of a stationary listener. The real freq. of whistle is 400 Hz. Calculate the apparant freq. heared by listner (a) when the engine is approaching the listener. (b) when the engine moving away from the listner (v_{sound} = 340 m/sec)

(A) 484 Hz, 340 Hz

(B) 220 Hz, 180 Hz

(C) 320 Hz,155 Hz

(D) 400 Hz, 330 Hz

Q16. A train moving towards a hill at a speed of 72 km / hr sounds a whistle of frequency 500 Hz. A wind is blowing from the hill at a speed of 36 km / hr. If the speed of sound in air is 340 m / s, the frequency heard by a man on the hill is

(A) 532.25 Hz.

(B) 565.0 Hz.

(C) 516.1 Hz.

(D) none of the above.

Q17. A source of sound of frequency 600 Hz is placed inside water. The speed of sound in water is 1500 m/s and in air it is 300 m/s. The frequency of sound recorded by an observer who is standing in air is :-

(A) 200 Hz (B) 3000 Hz

(C) 120 Hz (D) 600 Hz

Applications of Doppler's Effect

Q18. A man standing between two cliffs claps his hands and starts hearing a series of echoes at intervals of 2 sec. since the speed of sound in air is 340 m/sec. The distance between the cliffs must be:

(A) 340 m (B) 680 m

(C) 1020 m (D) 170 m

Interference

Q19. The ratio of maximum to minimum intensity due to superposition of two waves is $\frac{49}{9}$. Then the ratio of the intensity of component waves is :-

(A) $\frac{25}{4}$ (B) $\frac{16}{25}$

(C) $\frac{4}{49}$ (D) $\frac{9}{49}$

Q20. The energy in the superposition of waves :

(A) Is lost

(B) Increase

(C) remain same, only redistribution occurs

(D) None of the above

Q21. Two waves of intensities ratio are 9:1 then the ratio of their maximum and minimum intensities will be :-

(A) 10 : 8 (B) 7 : 2

(C) 4 : 1 (D) 2 : 1

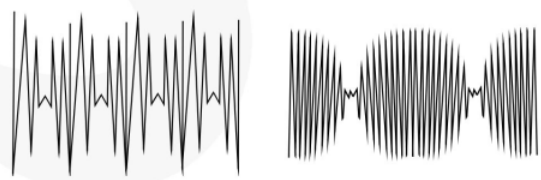
Beats

Q22. A tuning fork A produces 4 beats/sec with another tuning fork B of frequency 288 Hz. If fork is loaded with little wax. No of beats per sec decreases. The frequency of the fork A, before loading is -

(A) 290 Hz (B) 288 Hz

(C) 292 Hz (D) 284 Hz

Q23. Two waves with similar frequencies are added. The resulting waveform oscillates with the average frequency and with an oscillating amplitude that changes with a frequency equal to the difference between the original frequencies. These oscillations in the amplitude are known as beats. The traces show the resulting waveforms that occur when two different pairs of waves are added. Graph is for the same time interval in both cases, which of the following statements is TRUE?



(A) On average, the waves on the left had higher frequencies, but the difference between frequencies less

(B) On average, the waves on the right had higher frequencies, but the difference between frequencies less

(C) On average, the waves on the left had higher frequencies, but the difference between frequencies more

(D) On average, the waves on the right had higher frequencies, but the difference between frequencies more

Exercise - 1A — Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements are wrong about the velocity of sound in air:
(A) decreases with increases in temperature
(B) increases with decrease in temperature
(C) decreases as humidity increases
(D) independent of density of air

Q2. Three coherent source kept along the same line produce intensity I_0 each at point P on this line. When S_1 & S_2 are switched on simultaneously, intensity at point P is $2I_0$. When S_2 and S_3 are switched on simultaneously, intensity at point P is $2I_0$. Then -
(A) When S_1 and S_3 are switched on simultaneously, intensity at point P can be $2I_0$
(B) When S_1 and S_3 are switched on simultaneously, intensity at point P can be 0
(C) When all 3 sources are switched on simultaneously, intensity at point P can be I_0
(D) When all 3 sources are switched on simultaneously, intensity at point P can be $3I_0$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q3. Third overtone of a closed organ pipe is in unison with fourth harmonic of an open organ pipe. find the ratio of lengths of the pipes.

Q4. When both source and observer approach each other with a speed equal to the half the speed of sound, then determine the percentage change in frequency of sound as detected by the listener.

Q5. During interference phenomenon of two wave, it is observed that amplitude ratio of waves is 5 : 4. Find the maximum to minimum intensity ratio of resulting wave.

Q6. Two sound waves emitted by one sound source is approaching at a point through two paths. When path difference is 36 cm or 60 cm, the points are silence point. If speed of sound in air be 330 m/sec, then find the frequency of sound source.

SUBJECTIVE TYPE

Q7. Sound can be heard at great distances after rainfall. Explain.

Q8. A sound wave is propagating in given gaseous medium then plot graph between
(A) v_{gas} & T (B) v_{gas} & P
(C) v_{gas} & $\sqrt{T}$
[v_{gas} = Speed of sound in gas,
P = Pressure,
T = Temperature]
Also write down proper units on axis.

Q9. Use the formula $v = \sqrt{\frac{\gamma P}{\rho}}$ to explain why the speed of sound in air
(A) is independent of pressure.
(B) increases with temperature.
(C) increases with humidity.

Q10. A stone dropped from the top of a tower of height 300 m high splashes into the water of a pond near the base of the tower. When is the splash heard at the top, given that the speed of sound in air is 340 m s^{-1} ?
($g = 9.8\ ms^{-2}$)

Q11. A 20 cm long pipe is closed at one end. Which harmonic mode of the pipe is resonantly excited by 425Hz source ? Will the same source be in resonance with the pipe if both ends are open ?
(Speed of sound in air = 340 m/s)

Q12. A railway engine moving with a speed of 60 m/s passes a stationary listener. The real frequency of its whistle is 400 Hz. Calculate the apparent frequency heard by the listener.
(A) When the engine is approching the listener.
(B) When the engine is moving away from the listener.
(velocity of sound = 340 m/s)

Q13. A stationary source emits sound of frequency 1200 Hz. If wind blows at the speed of 0.1v, deduce
(A) The change in the frequency for a stationary observer on the wind side of the source.
(B) Report the calculations for the case when there is no wind but the observer moves at 0.1v speed towards the source.
(given : velocity of sound = v)

Q14. Two waves having same frequency and same amplitude are superimposed at a point. Find the phase difference between two waves if resultant amplitude is :
(A) 2a (B) $\sqrt{2}a$
(C) a (D) Zero

Q15. A Standard tuning fork contains frequency n_0, another TF n_A have 20% more compared to n_0, n_B 30% more compared to n_0. n_A & n_B have beat relation of 6 beat/sec, then determine n_0, n_A, n_B.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A firecracker exploding on the surface of a lake is heard as two sounds a time interval t apart by a man on a boat close to water surface. Sound travels with a speed u in water and a speed v in air. The distance from the exploding firecracker to the boat is

(A) $\frac{uvt}{u+v}$ (B) $\frac{t(u+v)}{uv}$

(C) $\frac{t(u-v)}{uv}$ (D) $\frac{uvt}{u-v}$

Q2. The speed of longitudinal wave is 100 times the speed of transverse wave in a taut brass wire. If the Young's modulus of brass is $1.0 \times 10^{11}\ N/m^2$, the stress in wire is :-
(A) $1.0 \times 10^{7}\ N/m^2$
(B) $1.0 \times 10^{6}\ N/m^2$
(C) $1.0 \times 10^{5}\ N/m^2$
(D) $1.0 \times 10^{8}\ N/m^2$

Q3. A is singing a note and at the same time B is also singing a note with 1/8th the frequency of A. The energies of the two sounds are equal. The displacement amplitude of the note of B is :
(A) same as that of A
(B) twice that of A
(C) four times that of A
(D) eight times that of A

Q4. Spherical sound waves are emitted uniformly in all directions from a point source. The variation in sound level SL as a function of distance r(> 0) from the source can be written as :-

(A) $SL = -b \log r^a$ (B) $SL = a - b \log r$

(C) $SL = a - b(\log r)^2$ (D) $SL = a - \frac{b}{r^2}$

Q5. The ratio of intensities between two coherent sound sources is 4 : 1. The difference of loudness in dB between maximum and minimum intensities, when they interfere in space is:

(A) 10 log 2 (B) 20 log 3

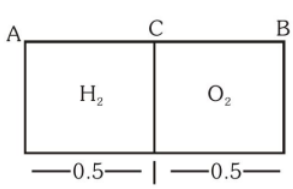
(C) 10 log 3 (D) 20 log 2

Q6. An open pipe is suddenly closed with the result that the second overtone of the closed pipe is found to be higher in frequency by 100 Hz, than the first overtone of the original pipe. Then the fundamental frequency of open pipe will be :

(A) 100 Hz (B) 300 Hz

(C)150 Hz (D) 200 Hz

Q7. As shown in fig AB is one metre long cylinder. At the ends A&B and middle point C there are thin flexible diaphragm. There is H_2 in AC and O_2 filled in part BC. diaphragm A & B are oscillated with some freq. When there is antinode at C then ratio of their minimum frequency n_{H_2} / n_{O_2} will be: (V_{H_2} = 1100 m /sec), V_{O_2} = 300 m/sec



(A) 11/3 (B) 11/7 (C) 9/7 (D) 9/11

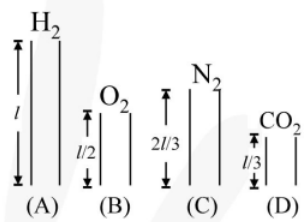
Q8. A closed organ pipe of length 1.5 m have some gas in it vibrating in fandamantal mode. Another open organ pipe of same length but filled with air, resonates with the same fork in its fundamental mode. If the temperature of the room be 30°C and the speed of sound in air at 30°C be 360 m/s, then speed of sound in the gas at 0°C is :

(A) 637 m/s (B) 683 m/s

(C) 341.5 m/s (D) 318 m/s

Q9. Four open organ pipes of different lengths and different gases at same temperature as shown in figure. Let f_A, f_B, f_C and f_D be their fundamental frequencies then :

[Take $\gamma_{CO_2} = 7/5$]



(A) $f_A/f_B = \sqrt{2}$ (B) $f_B/f_C = \sqrt{72/28}$

(C) $f_C/f_D = \sqrt{11/28}$ (D) $f_D/f_A = \sqrt{76/11}$

Q10. A cylindrical tube (L = 120 cm.) is resonant with a tuning fork of frequency 330 Hz. If it is filled by water then to get resonance minimum length of water column is-
(V_{air} = 330 m/s)

(A) 45 cm. (B) 60 cm.

(C) 25 cm. (D) 20 cm.

Q11. In a resonance tube experiment, an 80 cm air column is in resonance with a turning fork in first overtone. Which equation can represent correct pressure variation in the air column (x = 0 is the top point of the tube, neglect end correction, speed of sound = 320 m/sec) :-

(A) $A \sin \frac{15\pi}{8} x \cos 600\pi t$

(B) $A \cos \frac{15\pi}{8} x \sin 600\,\pi t$

(C) $A \cos \frac{15\pi}{8} x \sin 300\,\pi t$

(D) $A \sin \frac{15\pi}{8} x \sin 300\,\pi t$

Q12. In the experiment for the determination of the speed of sound in air using the resonance column method, the length of the air column that resonates in the fundamental mode, with a tuning fork is 0.1 m. When this length is changed to 0.35 m, the same tuning fork resonates with the first overtone. Calculate the end correction :-

(A) 0.012 m (B) 0.025 m

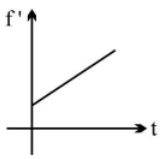
(C) 0.05 m (D) 0.024 m

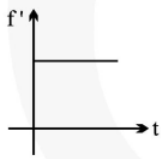
Q13. A source when at rest in a medium produces waves with a velocity v and a wavelength of λ . If the source is set in motion with a velocity v_s, what would be the wavelengths produced directly in front of the source?

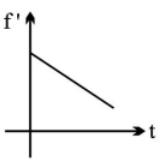
(A) $\lambda\left(1-\frac{v_s}{v}\right)$ (B) $\lambda\left(1+\frac{v_s}{v}\right)$

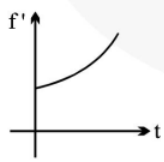
(C) $\lambda\left(1+\frac{v}{v_s}\right)$ (D) $\frac{\lambda v}{v+v_s}$

Q14. An observer starts moving with uniform acceleration 'a' towards a stationary sound source of frequency f. As the observer approaches the source, the apparent frequency f' heard by the observer varies with time t as :

(A) 

(B) 

(C) 

(D) 

Q15. An observer moves towards a stationary source of sound with a velocity one tenth the velocity of sound. The apparent increases in frequency is -

(A) zero (B) 5%

(C) 10% (D) 0.1%

Q16. A point source of sound is located somewhere along the x-axis. Experiments show that the same wave front simultaneously reaches listeners at x = –8 m and x = +2.0 m.

A third listener is positioned along the positive y-axis. What is her y-coordinate (in m) if the same wave front reaches her at the same instant as it does the first two listeners ?

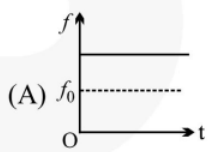
(A) 4 (B) 3 (C) 2 (D) 5

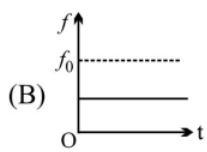
Q17. A source of sound S having frequency f. Wind is blowing from source to observer O with velocity u. If speed of sound with respect to air is C, the wavelength of sound detected by O is:

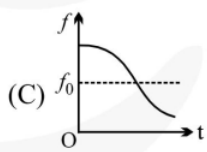
(A) $\frac{C+u}{f}$ (B) $\frac{C-u}{f}$

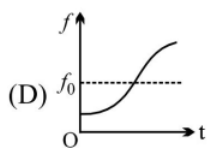
(C) $\frac{C(C+u)}{(C-u)f}$ (D) $\frac{C}{f}$

Q18. Source and observer both start moving simultaneously from origin, one along x-axis and the other along y-axis with speed of source = twice the speed of observer. The graph between the apparent frequency observed by observer f and time t would approximately be :

(A) 

(B) 

(C) 

(D) 

Q19. A siren placed at a railway platform is emitting sound of frequency 5 kHz. A passenger sitting in a moving train A records a frequency of 5.5 kHz while the train approaches the siren. During his return journey in a different train B he records a frequency of 6.0 kHz while approaching the same siren. The ratio of the velocity of train B to that of train A is :-

(A) 242/252 (B) 2

(C) 5/6 (D) 11/6

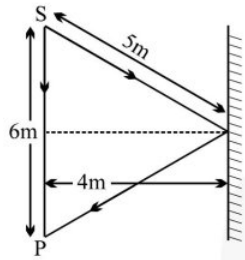
Q20. A person standing at a distance of 6 m from a source of sound receives sound wave in two ways, one directly from the source and other after reflection from a rigid boundary as shown in the figure. The maximum wavelength for which, the person will receive maximum sound intensity, is

(A) 4 m

(B) $\frac{16}{3}$ m

(C) 2 m

(D) $\frac{8}{3}$ m



Q21. A source is moving with constant speed $v_s = 20$ m/sec towards a stationary observer due east of source. Wind is blowing at the speed of 20 m/sec due to 60° north of east. The source is generating of frequency 500 Hz. Then frequency registered by observer is :
[Speed of sound in still air = 330 m/sec.]

(A) 500 Hz (B) 532 Hz

(C) 531 Hz (D) 530 Hz

Q22. The equations of S.H.M. of medium particle due to sound waves propagating in a medium are given by $s_1 = 2 \sin(200\pi t)$ and $s_2 = 5 \sin(150\pi t)$. The ratio of average intensities of sound at these points is :

(A) 4 : 25 (B) 9 : 100

(C) 8 : 15 (D) 64 : 225

Q23. Two waves of wave length 2 m and 2.02 m respectively moving with the same velocity superimpose to produce 2 beats per sec. The velocity of the wave is:

(A) 400.0 m/s (B) 402 m/s

(C) 404 m/s (D) 406 m/s

Q24. Frequency of tuning fork A is 256 Hz. It produces 4 beats/sec with tuning fork B. When wax is applied at tuning fork B then 6 beats/sec are heard. By reducing little amount of wax 4 beats/sec are heard. Frequency of B is :

(A) 250 Hz (B) 260 Hz

(C) 252 Hz (D) 256 Hz

Q25. Two wires are fixed in a sonometer. Their tensions are in the ratio 8:1. The lengths are in the ratio 36:35. the diameters are in the ratio 4:1. Densities of the materials are in the ratio 1:2. If the higher frequency in the setting is 360 Hz, the beat frequency when the two wires sounded together is :

(A) 8 (B) 5 (C) 10 (D) 6

Q26. A person is observing two trains. One of them is coming with speed of 4m/sec. and another is going with same speed. If two trains blowing a whistle with frequency 240 Hz. The beat frequency heard by stationary person will be (speed of sound in air = 320 m/sec.)

(A) zero (B) 3 (C) 6 (D) 12

Q27. Two vibrating tuning forks produce progressive waves given by $y_1 = 4 \sin(500\pi t)$ and $y_2 = 2 \sin(506\pi t)$. These tuning forks are held near the ear of a person. The person will hear :

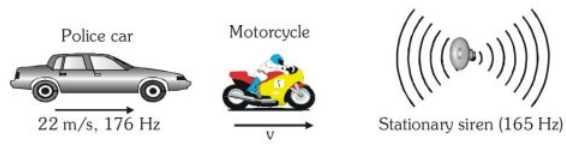
(A) 3 beats/s with intensity ratio between maxima and minima equal to 4.

(B) 3 beats/s with intensity ratio between maxima and minima equal to 9.

(C) 6 beats/s with intensity ratio between maxima and minima equal to 4.

(D) 6 beats/s with intensity ratio between maxima and minima equal to 9.

Q28. A police car moving at 22 m/s chases a motocyclist. The police man sounds his horn at 176 Hz, while both of them move towards a stationary siren of frequency 165 Hz. Calculate the speed of the motorcycle. If it is given that the motorcyclist does not observe any beats :–



(A) 33 m/s (B) 22 m/s

(C) zero (D) 11 m/s

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A sound wave has a wavelength of 3.0 m. The distance from a compression center to the adjacent rarefaction center is :-

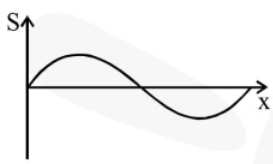
(A) 0.75 m

(B) 1.5 m

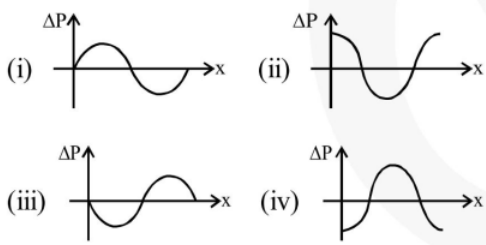
(C) 3.0 m

(D) need to know wave speed

Q2. If a sound wave is travelling and snap shot at t = 0 is as shown in figure.



Choose snapshot of pressure variation.



(A) For wave travelling towards right or left (i) is correct.

(B) For wave travelling towards right graph (iv) and for wave travelling towards left graph (iv) is correct.

(C) For wave travelling towards right graph (i) and for wave travelling towards left graph (iii) is correct.

(D) For wave travelling towards right or left (ii) is correct.

Q3. The particle displacement of a travelling longitudinal wave is represented by S = S (x, t). The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true?

(A) $|\partial S / \partial x|_C = |\partial S / \partial x|_R$

(B) $|\partial S / \partial t|_C = |\partial S / \partial t|_R = 0$

(C) $(\text{pressure})_C - (\text{pressure})_R = 2\,|\partial S / \partial x|_C \times$ Bulk modulus of air.

(D) Particles of air are stationary mid-way between 'C' and 'R'.

Q4. Which of the following is the equation of a spherical wave :-

(A) $S = S_0 \sin(Kx - \omega t)$

(B) $S = S_0 \cos(Kx - \omega t)$

(C) $S = (S_0/x) \sin(\omega t - Kx)$

(D) $S = (S_0/x^2) \sin(\omega t - Kx)$

Q5. A note is produced when you blow air across the top of a test tube. Two students were asked about the effect of blowing harder.

Student-A : The pitch of sound would increase.

Student-B : The intensity of sound would increase.

(A) A is correct, B is wrong

(B) B is correct, A is wrong

(C) both are correct

(D) both are wrong

Q6. A person hears a single-frequency sound from a speaker that emits sound uniformly in all directions. The person records the decibel level reading. Which of the following processes would increase the decible reading the most ?

(A) The power output of the speaker is doubled.

(B) The frequency of the sound is doubled keeping amplitude constant.

(C) The temperature of the air is doubled on the Kelvin scale.

(D) The area over which the sound is transmitted is halved.

Q7. An open pipe is in resonance in 2nd harmonic with frequency f_1. Now one end of the tube is closed and frequency is increased to f_2 such that the resonance again occurs in nth harmonic. Choose the correct option :-

(A) $n = 3, f_2 = \frac{3}{4} f_1$ (B) $n = 3, f_2 = \frac{5}{4} f_1$

(C) $n = 5, f_2 = \frac{5}{4} f_1$ (D) $n = 5, f_2 = \frac{3}{4} f_1$

Q8. A sound consists of four frequencies : 300 Hz, 600 Hz, 1200 Hz and 2400 Hz. A sound 'filter' is made by passing this sound through a bifurcate pipe as shown. The sound wave has to travel a distance of 50 cm more in the right branch-pipe than in the straight pipe. The speed of sound in air is 300 m/s. Then, which of the following frequencies will be almost completely muffled or 'silenced' at the outlet?



(A) 300 Hz (B) 600 Hz
(C) 1200 Hz (D) 2400 Hz

Q9. Two tuning forks of frequency 250 Hz and 256 Hz produce beats. If a maximum of intensity is observed just now, after how much time the minimum is observed at the same place ?

(A) $\frac{1}{18}$ sec (B) $\frac{1}{4}$ sec.

(C) $\frac{1}{3}$ sec. (D) $\frac{1}{12}$ sec.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 10 to 12

A metallic rod of length 1m has one end free and other end rigidly clamped. Longitudinal stationary waves are set up in the rod in such a way that there are total six antinodes present along the rod. The amplitude of an antinode is 4×10^{-6} m. Young's modulus and density of the rod are 6.4×10^{10} N/m^2 and 4×10^3 Kg/m^3 respectively. Consider the free end to be at origin and at t=0 particles at free end are at positive extreme.

Q10. The equation describing displacements of particles about their mean positions is -

(A) $s = 4\times10^{-6}\cos\left(\frac{11\pi}{2}x\right)\cos(22\pi\times10^3 t)$

(B) $s = 4\times10^{-6}\cos\left(\frac{11\pi}{2}x\right)\sin(22\pi\times10^3 t)$

(C) $s = 4\times10^{-6}\cos(5\pi x)\cos(20\pi\times10^3 t)$

(D) $s = 4\times10^{-6}\cos(5\pi x)\sin(20\pi\times10^3 t)$

Q11. The equation describing stress developed in the rod is -

(A) $140.8\pi\times10^4\cos\left(\frac{11}{2}\pi x+\pi\right)\cos(22\pi\times10^3 t)$

(B) $140.8\pi\times10^4\sin\left(\frac{11}{2}\pi x+\pi\right)\cos(22\pi\times10^3 t)$

(C) $128\pi\times10^4\cos(5\pi x+\pi)\cos(20\pi\times10^3 t)$

(D) $128\pi\times10^4\sin(5\pi x+\pi)\cos(20\pi\times10^3 t)$

Q12. The magnitude of strain at midpoint of the rod at t = 1 sec is -

(A) $11\sqrt{3}\pi\times10^{-6}$

(B) $11\sqrt{2}\pi\times10^{-6}$

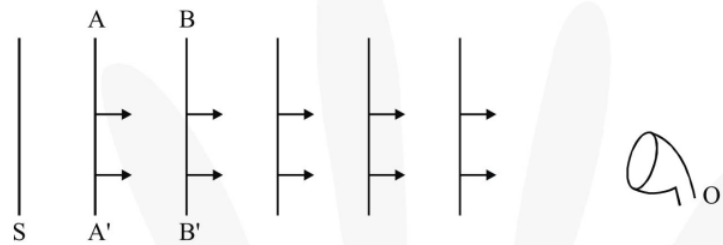
(C) $10\sqrt{3}\pi\times10^{-6}$

(D) $10\sqrt{2}\pi\times10^{-6}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13. Consider a large plane diaphragm 'S' emitting sound and a detector 'O'. The diagram shows plane wavefronts for the sound wave travelling in air towards right when source, observer and medium are at rest. AA' and BB' are fixed imaginary planes. Column-I describes about the motion of source, observer or medium and column-II describes various effects. Match them correctly.



Column-I	Column-II
(A) Source starts moving towards right	(P) Distance between any two wavefronts will increase.
(B) Air starts moving towards right	(Q) Distance between any two wavefronts will decrease.
(C) Observer and source both move towards left with same speed.	(R) The time needed by sound to move from plane AA' to BB' will increase.
(D) Source and medium (air) both move towards right with same speed.	(S) The time needed by sound to move from plane AA' to BB' will decrease.
	(T) Frequency received by observer increases.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q14. A bat is flitting about in a cave, navigating via ultrasonic bleeps. Assume that the sound emission frequency of the bat is 40 kHz. During one fast swoop directly toward a flat wall surface, the bat is moving at 0.03 times the speed of sound in air. What frequency does the bat hear reflected off the wall ?

SUBJECTIVE TYPE

Q15. Two narrow cylindrical pipes A and B have the same length. Pipe A is open at both ends and is filled with a monoatomic gas of molar mass M_A. Pipe B is open at one end and closed at the other end, and is filled with a diatomic gas of molar mass M_B. Both gases are at the same temperature.

(A) If the frequency of the second harmonic of the fundamental mode in pipe A is equal to the frequency of the third harmonic of the fundamental mode in pipe B, determine the value of M_A/M_B.

(B) Now the open end of pipe B is also closed (so that the pipe is closed at both ends). Find the ratio of the fundamental frequency in pipe A to that in pipe B.

Q16. A steel rod having a length of 1 m is fastened at its middle. Assuming young's modulus to be 2×10^{11} Pa and density to be 8 gm/cm^3, find the fundamental frequency of the longitudinal vibration and frequency of first overtone.

Q17. A car is moving towards a huge wall with a speed = c/10 , where c = speed of sound in still air. A wind is also blowing parallel to the velocity of the car in the same direction and with the same speed. If the car sounds a horn of frequency f, then what is the frequency of the reflected sound of the horn heard by driver of the car?

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
C	B	B	D	A	B	D	A	C	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	A	A	B	A	A	D	C	A	C
21	**22**	**23**							
C	C	B							

Ex-1A

1. A,B,C,D **2.** B,C **3.** 7/8 **4.** 200% **5.** 81:1

6. 1375 Hz **7.** Explainatory Question **8.** Explainatory Question

9. Explainatory Question **10.** 8.707 s **11.** first harmonic, NO

12. (A) 485.7 Hz, (B) 340 Hz **13.** (A) No change in frequency (B) 1320 Hz

14. (A)0°, (B)90°, (C)120°, (D)180° **15.** 60 Hz, 72 Hz, 78 Hz

Ex-2

1	2	3	4	5	6	7	8	9	10
D	A	D	B	B	D	A	B	C	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	B	A	A	C	A	A	B	B	A
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**		
C	D	C	B	C	C	B	B		

Ex-2A

1. B **2.** B **3.** A,C,D **4.** C **5.** B

6. B **7.** C **8.** A **9.** B,D **10.** A

11. B **12.** B **13.** (A) (Q,T) ; (B) (P,S) ; (C) (P) (D) (S,T)

14. 42.47 Hz **15.** (A) 2.116, (B) $\frac{3}{4}$ **16.** 2.5 kHz, 7.5 kHz

17. 11f / 9